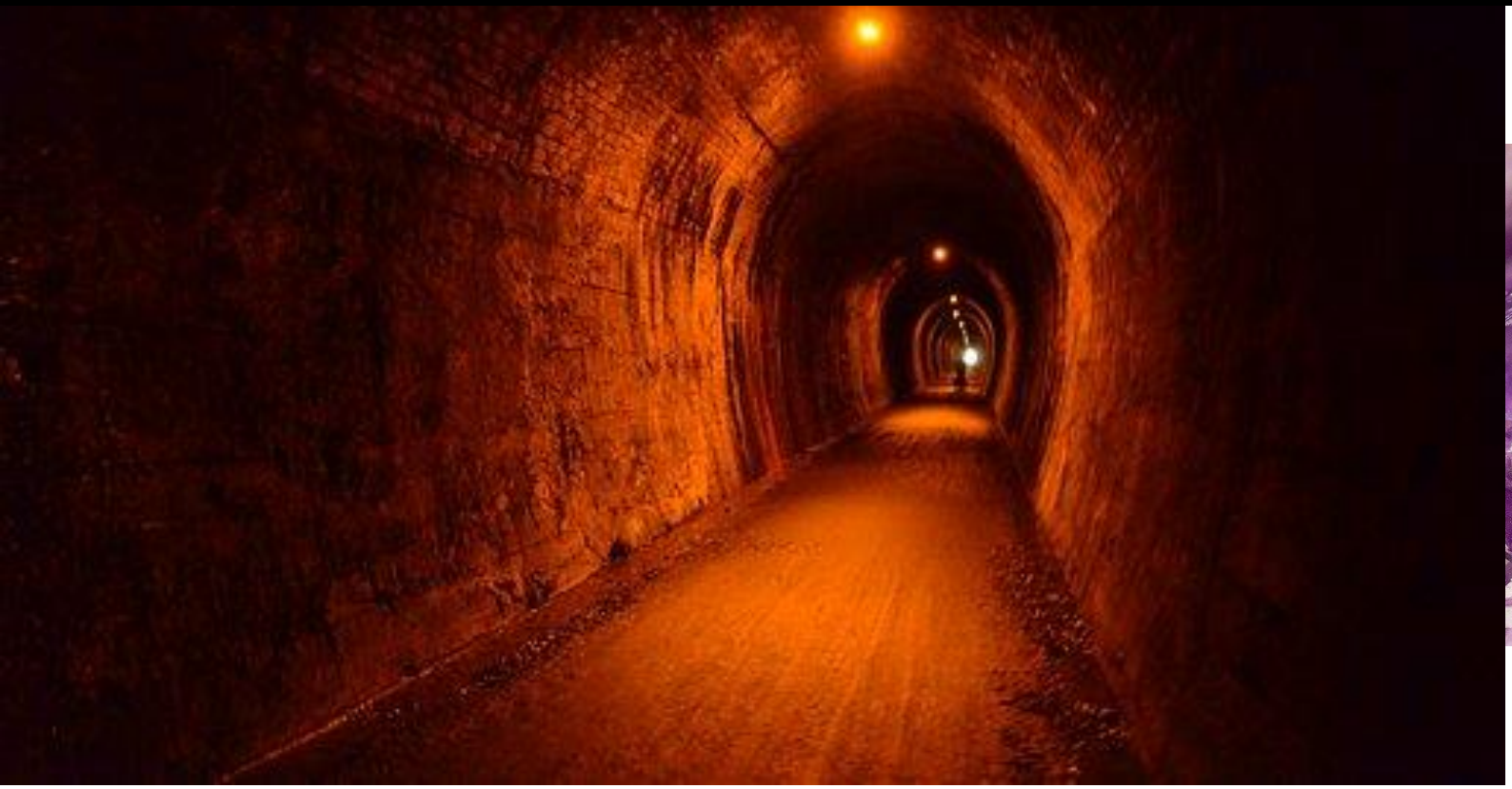
EART11200: Natural Scientist's Toolkit





Before looking at the mining problem we will consider Bragg's law. This helps determine the distance between planes in a crystal, and uses trigonometry.

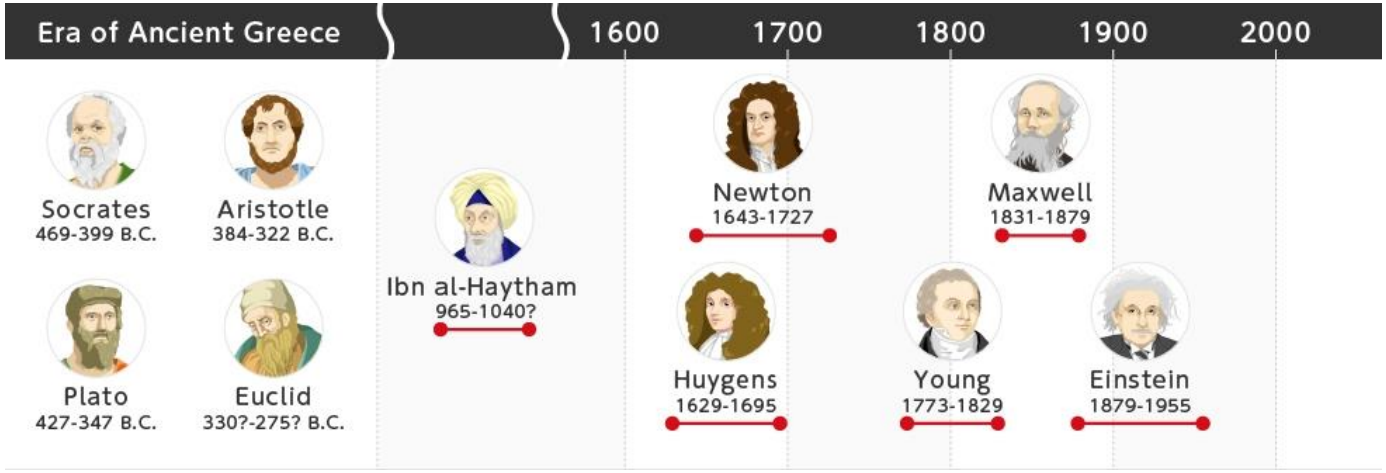
Langworthy professor of physics at Manchester
1919 - 1937

With father William Henry Bragg shared the Nobel prize for physics in 1915, *"for their services in the analysis of crystal structure by means of X-rays".*

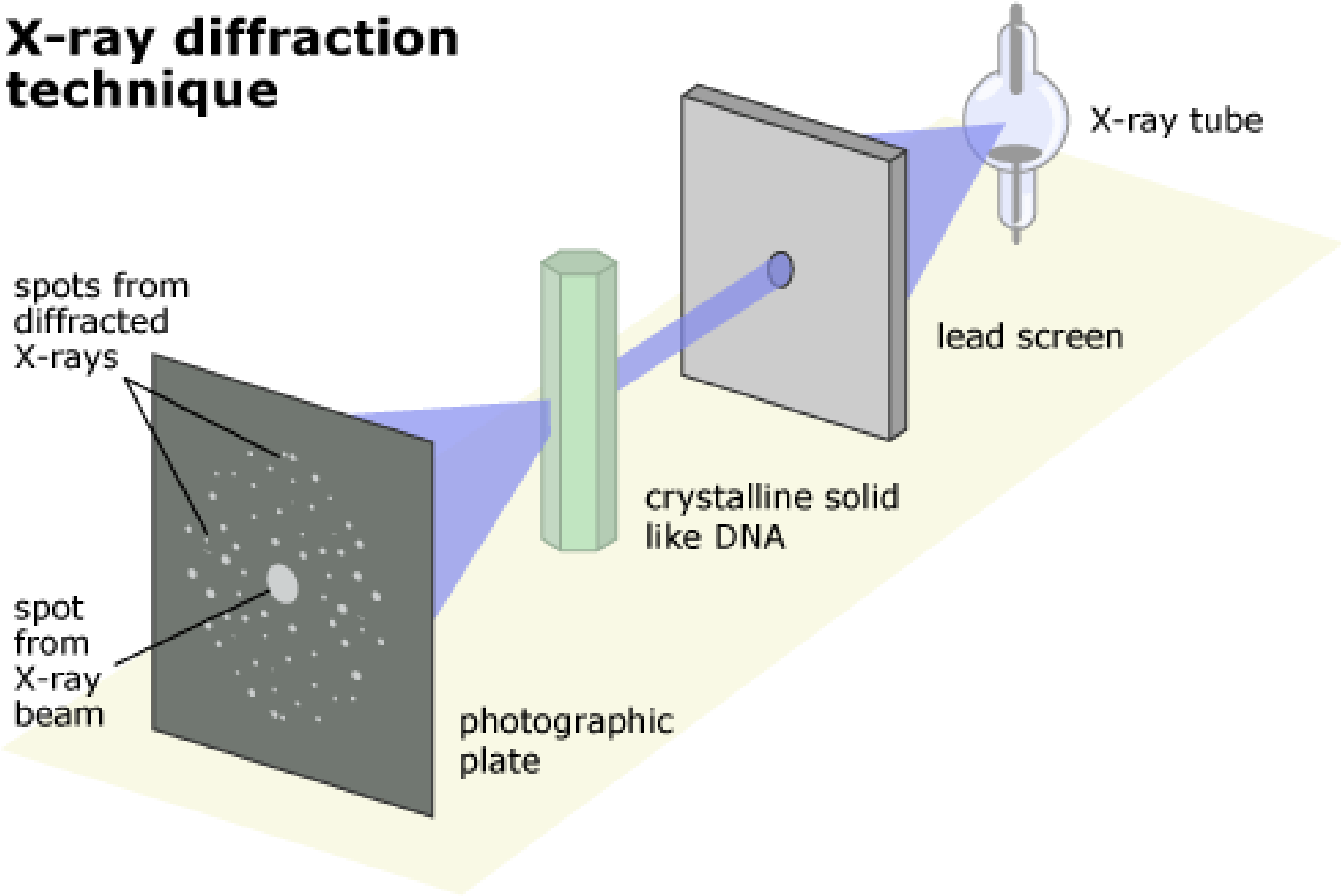


William Lawrence Bragg

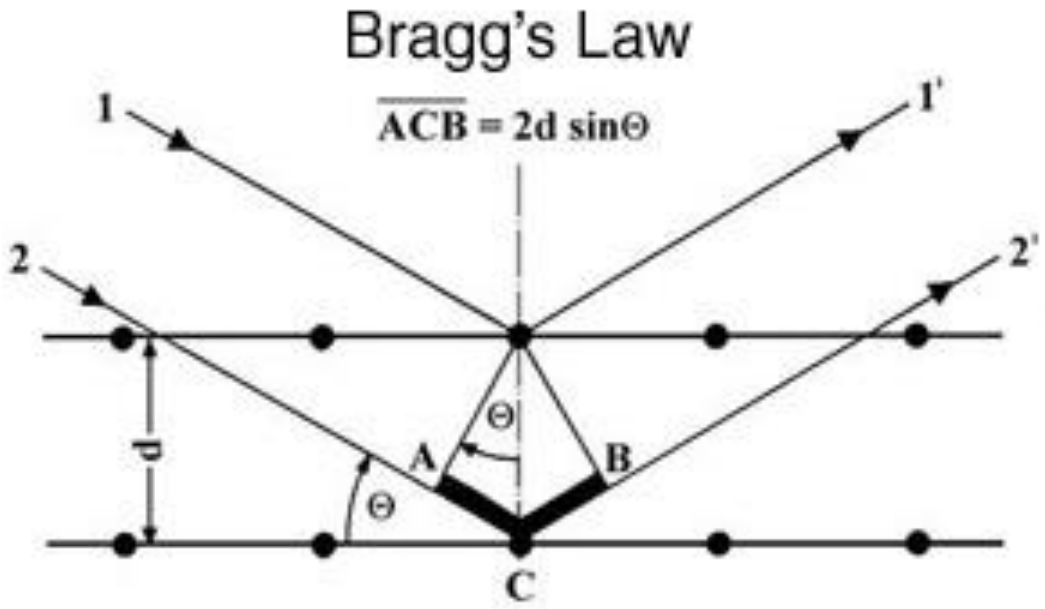
Still widely believed that light was a "wave" (a disturbance in the electro-magnetic field).

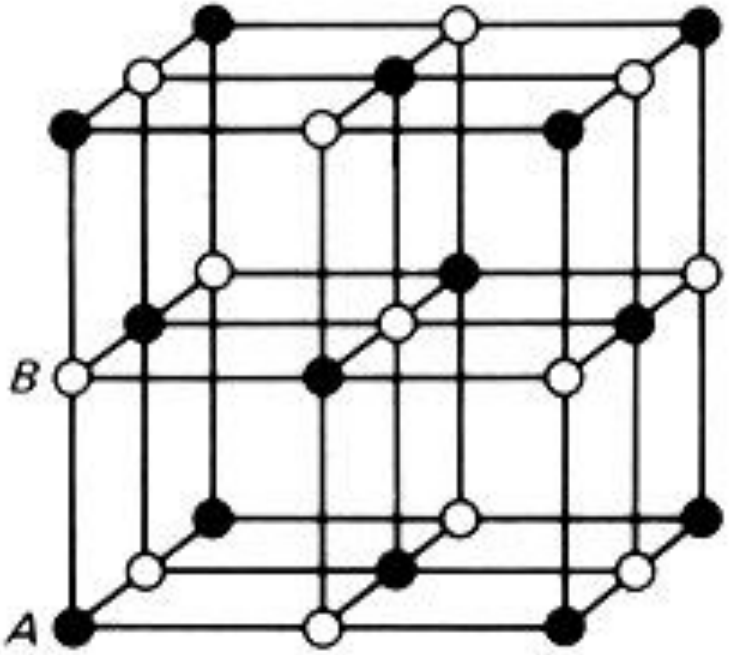


http://photonterrace.net/en/photon/history/



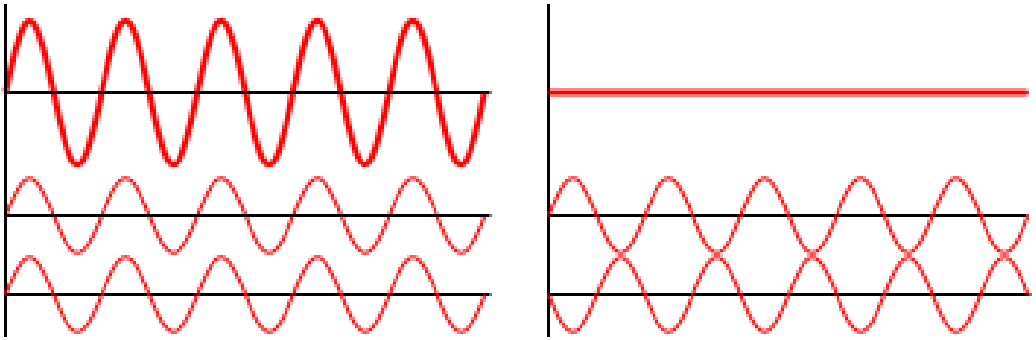
Visual of how X-Ray diffraction works © King's College Archives; Wilkins-Gosling





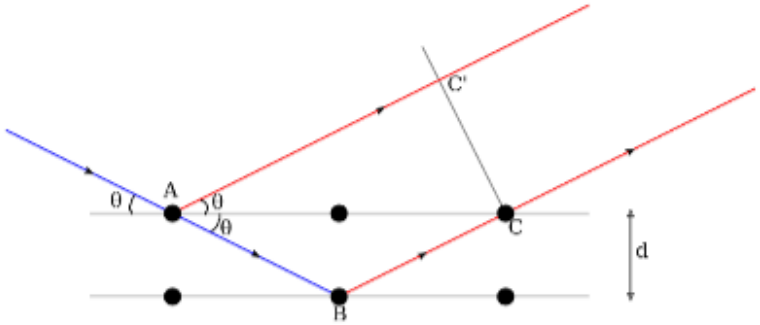
To understand Bragg's law, consider light as a wave.

Constructive interference



Destructive interference

original version: Haade;vectorization: Wjh31, Quibik / CC BY-SA (https://creativecommons.org/licenses/by-sa/3.0)



The arrangement of atoms in the crystal above shows the principle behind Bragg's law. The incoming ray, at angle θ, is scattered at angle θ off the first layer of atoms. The same is true for the next layer. We consider light as a wave here (with troughs and crests like the sea). For some special angle, θ the two scattered rays will either completely cancel each other out or completely add together (constructive interference) in the strongest way possible. For constructive interference the distance: $\vec{AB} + \vec{BC} - \vec{AC'}$ must equal an integer number of wavelengths, λ.
The distance $\vec{AB} = \vec{BC}$ is equal to $\frac{d}{\sin\theta}$, whereas $\tan\theta = \frac{d}{\frac{1}{2}\vec{AC}}$.
The distance $\vec{AC}$ is therefore $\frac{2d}{\tan\theta}$. The distance $\vec{AC'}$ is $\vec{AC}$ multiplied by $\cos\theta$.
So we write down $n\lambda = \vec{AB} + \vec{BC} - \vec{AC'}$:

$$\begin{aligned} n\lambda &= \frac{2d}{\sin\theta} - \frac{2d}{\tan\theta}\cos\theta \\ &= \frac{2d}{\sin\theta} \times \sin^2\theta \\ &= 2d \times \sin\theta \end{aligned}$$

$\tan\theta = \sin\theta / \cos\theta$
$1-\cos^2\theta = \sin^2\theta$

(where we have used $1 \equiv \sin^2\theta + \cos^2\theta$) which is known as Bragg's law.

Example problem: For a crystal lattice with layers separated by a distance d = 50Å, what angles, measured from the face of the lattice, will constructively interfere if the incident radiation is X-rays of wavelength 1 nm?

Use Bragg's law, and re-arrange the equation to give an expression for sinθ:

$$\sin\theta = \frac{n\lambda}{2d}$$

Substitute our values in to get:

$$\sin\theta = \frac{n \times 1 \times 10^{-9}}{2 \times 50 \times 10^{-10}} = 0.1n$$

with $n = 1, 2, 3, \ldots$

find the angles by taking inverse sin *to get* $\theta = 5.74°, 11.54°, 17.46° \ldots$.

1.4.4.2 Problem

1. Bragg's law: you study a crystal with X-rays of wavelength 2 nm and note the destructive interference minimum in intensity occurs at an angle of $\theta = 10°$. Use Bragg's law to estimate the separation distance between layers in the crystal.

This is one of the problems for the practical this week, so no need to complete it now.